

Paper 2 - Set 06 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	B	Circles and circle theorems
2	C	Differentiation
3	C	Quadratic functions
4	E	Exponentials and logarithms
5	D	The logic of arguments
6	F	Sequences and series
7	A	Identifying errors in proofs
8	A	Mathematical proof
9	B	Angles, polygons, congruence and vectors
10	C	Exponentials and logarithms
11	A	Identifying errors in proofs
12	C	The logic of arguments
13	G	Ratio, proportion and percentages
14	D	Mathematical proof
15	C	Chains of reasoning
16	D	Chains of reasoning
17	H	Quadratic functions
18	D	Indices and surds
19	A	Identifying errors in proofs
20	A	The logic of arguments

Worked solutions

Question 1 CIR-03 · Circles and circle theorems · spec MM3.2

The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.

Answer: B

Radius direction $(3, 4)$ has gradient $\frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:

$$y - 3 = -\frac{3}{4}(x - 5) \Rightarrow 4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27.$$

Full method - how and why

A tangent is perpendicular to the radius at the point of contact. Find the radius gradient, invert, and write the line.

1 Vector from centre $(2, -1)$ to the point $(5, 3)$: $(3, 4)$, so the radius has gradient $\frac{4}{3}$.

why: Differences of coordinates give the direction; no need for the circle's equation.

2 The tangent is perpendicular, so its gradient is $-\frac{3}{4}$.

why: Tangent $\perp$ radius is the key circle fact here.

3 Line through $(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5)$.

why: The tangent touches at the given point, so use it in the point-gradient form.

4 Clear fractions: $4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27$.

why: Multiply by 4, then collect.

Where it goes wrong

- Using the radius gradient $\frac{4}{3}$ for the tangent, giving $4x - 3y = 11$ - an offered option.
- Arithmetic slip on the constant, giving $3x + 4y = 15$.

Takeaway Tangent gradient $= -1 \div (\text{radius gradient})$; the point of contact is on both the circle and the tangent.

Question 2 DIF-18 · Differentiation · spec MM6.3

Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.

Answer: C

Derivative $1 - \frac{4}{x^2} = 0$ at $x = 2$; value $2 + 2 = 4$ (second derivative $\frac{8}{x^3} > 0$, so a minimum).

Full method - how and why

Minimising $x + \frac{4}{x}$. Differentiate, solve, and confirm it is a minimum.

1 Write as powers: $y = x + 4x^{-1}$ and differentiate: $y' = 1 - 4x^{-2}$.

why: $\frac{d}{dx}(4x^{-1}) = -4x^{-2}$.

2 Solve $y' = 0$: $1 = \frac{4}{x^2} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root since $x > 0$).

why: The restriction $x > 0$ discards $x = -2$.

3 Confirm a minimum: $y'' = 8x^{-3} > 0$ for $x > 0$.

why: Positive second derivative means the curve bends upwards.

4 Evaluate: $y = 2 + \frac{4}{2} = 4$.

why: (Also follows from AM-GM: $x + \frac{4}{x} \geq 2\sqrt{4} = 4$.)

Where it goes wrong

- Giving the minimising $x = 2$ instead of the minimum value 4 - both are offered.
- Including $x = -2$, where the expression equals -4 but $x > 0$ is required.

Takeaway $x + \frac{k}{x} \geq 2\sqrt{k}$ for $x > 0$, with equality at $x = \sqrt{k}$ - worth recognising instantly.

Question 3 QUA-12 · Quadratic functions · spec MM1.3

The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .

Answer: C

$$b^2 - 144 = 0, \text{ so } b = \pm 12; \text{ their sum is } 0.$$

Full method - how and why

'Equal roots' is the discriminant-zero condition; the twist is that it gives two values of b and the question wants their sum.

- 1 Write the discriminant with $a = 2$, $c = 18$: $b^2 - 4(2)(18) = b^2 - 144$.

why: Substitute carefully - $4ac$ is 144, not 36.

- 2 Set it to zero: $b^2 = 144$, so $b = 12$ or $b = -12$.

why: Equal (repeated) roots means the discriminant vanishes; a square equal to a positive number always has two solutions.

- 3 Add them: $12 + (-12) = 0$.

why: The two values are symmetric about zero, so the sum is 0 without any further work.

Where it goes wrong

- Giving only the positive root and answering 12.
- Reading 'sum of the possible values of b ' as 'the value of b '.

Takeaway $b^2 = k$ always has two solutions $\pm\sqrt{k}$; read whether the question wants one of them, both, or a combination.

Question 4 LOG-16 · Exponentials and logarithms · spec MM5.1

For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

Answer: E

$$x = 0: 1 < 2 \text{ true. } x = 1: 3 < 4 \text{ true. } x = 2: 9 < 8 \text{ false. So I and II.}$$

Full method - how and why

Three numerical tests. Nothing conceptual - just careful arithmetic with small powers.

- 1 I: $x = 0$ gives $3^0 = 1$ and $2^1 = 2$. Is $1 < 2$? Yes. **True.**

why: $a^0 = 1$ for any a .

- 2 II: $x = 1$ gives $3^1 = 3$ and $2^2 = 4$. Is $3 < 4$? Yes. **True.**

why: Careful with the index $x + 1 = 2$.

- 3 III: $x = 2$ gives $3^2 = 9$ and $2^3 = 8$. Is $9 < 8$? No. **False.**

why: 3^x has overtaken 2^{x+1} by here - exponential growth with the larger base always wins eventually.

- 4 I and II only.

why: Match to the option list.

Where it goes wrong

- Computing 2^{x+1} as $2^x + 1$.
- Assuming the inequality holds for all x because it holds for the first two.

Takeaway Substitute each value carefully; a larger base always overtakes eventually, whatever the head start.

Question 5 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: ****if n is prime, then n is odd.**** Which one of the following is true?

Answer: D

$n = 2$ is an even prime, so the statement is false. The converse, 'if n is odd then n is prime', fails for $n = 9$.

Full method - how and why

Statement and converse again, with the classic $n = 2$ counterexample.

- 1 Statement: primes are odd? $n = 2$ is prime and even. **False.**
why: 2 is the only even prime - it is the standard counterexample and worth memorising.
- 2 Converse: odd numbers are prime? $n = 9 = 3 \times 3$ is odd and composite. **False.**
why: Also $n = 1$, which is odd and not prime by definition.
- 3 So both are false.
why: Match to the option list.

Where it goes wrong

- Forgetting that 2 is prime and declaring the statement true.
- Forgetting that 1 is **not** prime when hunting counterexamples.

Takeaway 2 is the even prime and 1 is not prime - the two facts behind most prime-related counterexamples.

Question 6 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

- I** $1 + 0.9 + 0.81 + \dots$
II $3 - 3 + 3 - 3 + \dots$
III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

A geometric series converges if and only if $|r| < 1$. I has $r = 0.9$, III has $r = 0.1$: both converge. II has $r = -1$ and does not converge.

Full method - how and why

Convergence of a geometric series depends only on $|r| < 1$. Compute r for each and apply the test.

- 1 I: $r = \frac{0.9}{1} = 0.9$, and $|0.9| < 1$: **converges.**
why: A ratio just below 1 still converges, though slowly.
- 2 II: $r = \frac{-3}{3} = -1$, and $|-1| = 1$ is not less than 1: **does not converge.**
why: The partial sums oscillate between 3 and 0 forever - they never settle, so there is no sum to infinity.
- 3 III: $r = \frac{10}{100} = 0.1$: **converges.**
why: Small ratio, rapid convergence.
- 4 I and III only.
why: Match the verdicts to the option list.

Where it goes wrong

- Thinking II converges to 0 or to 1.5 because the terms 'average out' - the partial sums must approach a single limit.
- Judging by the size of the terms rather than the ratio: III starts at 100 but converges fastest.

Takeaway Convergence is decided by $|r| < 1$ alone - not by the size of the first term or by whether the terms alternate.

Question 7 ERR-18 · Identifying errors in proofs · spec Err2

A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

Answer: A

Dividing by $\cos x$ loses $\cos x = 0$, i.e. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$: two solutions.

Full method - how and why

Dividing a trigonometric equation by $\cos x$ loses the solutions where $\cos x = 0$.

1 Line I divides by $\cos x$, valid only if $\cos x \neq 0$. **First error.**

why: The equation $2\cos^2 x = \cos x$ is satisfied whenever $\cos x = 0$, so those solutions are discarded.

2 The correct method: $2\cos^2 x - \cos x = 0$, so $\cos x(2\cos x - 1) = 0$.

why: Factorise rather than divide - it keeps both branches.

3 Branch $\cos x = 0$: in $0 \leq x < 2\pi$ that is $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ - **two** missed solutions.

why: Cosine vanishes at the top and bottom of the unit circle.

4 Branch $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$ - the two the student found.

why: Cosine is positive in the first and fourth quadrants, so $x = \frac{\pi}{3}$ and $2\pi - \frac{\pi}{3}$. Four solutions in total.

Where it goes wrong

- Saying one solution was missed; $\cos x = 0$ has two solutions in a full revolution.
- Blaming line III - the student's own two values are correct for their (incomplete) equation.
- Declaring the working correct because the two values found do satisfy the original equation.

Takeaway Never divide a trigonometric equation by a trig function - factorise, and count the solutions of **each** branch across the whole interval.

Question 8 PRF-20 · Mathematical proof · spec Prf3

T_n denotes the n th triangular number $\frac{n(n+1)}{2}$. A student observes $T_1 + T_2 = 4$, $T_2 + T_3 = 9$, $T_3 + T_4 = 16$ and conjectures that $T_n + T_{n+1}$ is always a perfect square. Which of the following proves the conjecture?

Answer: A

Only A is a general algebraic verification.

Full method - how and why

Proving a conjecture about a sequence: substitute the formula and simplify algebraically to a general result.

- Write both terms: $T_n = \frac{n(n+1)}{2}$ and $T_{n+1} = \frac{(n+1)(n+2)}{2}$.
why: Replace n by $n+1$ throughout the given formula.
- Add over the common denominator: $\frac{n(n+1) + (n+1)(n+2)}{2}$.
why: Same denominator, so just add numerators.
- Factor out $(n+1)$: $\frac{(n+1)[n + (n+2)]}{2} = \frac{(n+1)(2n+2)}{2}$.
why: Spotting the common factor avoids expanding and re-factorising.
- $2n+2 = 2(n+1)$, so the expression is $(n+1)^2$ - a perfect square for every n . **Proved.**
why: The 2 cancels the denominator.

Where it goes wrong

- Checking $T_4 + T_5 = 25$ and stopping - one more case is still not a proof.
- 'Triangular numbers are always squares' - plainly false ($T_2 = 3$).
- ' $T_n + T_{n+1} = 2T_n + (n+1)$, which is a square' - the identity is right but the conclusion is asserted, not shown.
- Quoting the difference $T_{n+1} - T_n = n+1$, which is true but says nothing about the sum.

Takeaway Substitute the general formula and simplify; consecutive triangular numbers sum to $(n+1)^2$.

Question 9 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.

Full method - how and why

A quadrilateral hierarchy question. Bisecting diagonals characterise parallelograms, a larger class than rectangles.

- Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other. So $Q \Rightarrow P$: ****P** is necessary.
why: Opposite sides equal and parallel forces the diagonals to share a midpoint.
- Is P sufficient? Take a non-rectangular parallelogram, e.g. a rhombus with angles $60^\circ/120^\circ$: diagonals bisect each other, but it is not a rectangle.
why: Bisecting diagonals say nothing about the angles.
- So P is necessary but not sufficient.
why: For a rectangle you also need the diagonals to be **equal** in length.

Where it goes wrong

- Answering 'necessary and sufficient' by confusing 'bisect each other' with 'bisect each other and are equal'.
- Getting the direction backwards: the broader property (parallelogram) is necessary for the narrower one (rectangle).

Takeaway Diagonals bisect each other $\Leftrightarrow$ parallelogram; add equal diagonals for a rectangle, add perpendicular for a rhombus.

Question 10 LOG-07 · Exponentials and logarithms · spec MM5.2

Which of the following statements is/are true for all positive real x and y ?

- I** $\log_{10}(x + y) = \log_{10} x + \log_{10} y$
II $\log_{10}(xy)^2 = 2\log_{10} x + 2\log_{10} y$
III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

Answer: C

II is correct: $\log(xy)^2 = 2\log(xy) = 2\log x + 2\log y$. I and III confuse the laws: log of a sum has no simple form, and a difference of logs is log of a quotient, not a quotient of logs.

Full method - how and why

Three log 'laws', two of which are fabrications. Test each against the genuine rules.

- 1** I: $\log(x + y)$ has no expansion. Test $x = y = 1$: $\log 2 \approx 0.30$ but $\log 1 + \log 1 = 0$. **False.**
why: The product law is $\log(xy) = \log x + \log y$; the sum inside is a different animal entirely.
- 2** II: $\log(xy)^2 = 2\log(xy) = 2(\log x + \log y) = 2\log x + 2\log y$. **True.**
why: Power law first, then product law - both applied correctly.
- 3** III: $\log x - \log y = \log \frac{x}{y}$, which is not the same as $\frac{\log x}{\log y}$. Test $x = 100$, $y = 10$: left = 1, right = $\frac{2}{1} = 2$. **False.**
why: Difference of logs is the log of a quotient, not the quotient of logs.
- 4** II only.
why: Match to the options.

Where it goes wrong

- Accepting I by analogy with the product law.
- Accepting III because 'subtraction corresponds to division' - it does, but inside the log.

Takeaway The laws act on what is **inside** the log: $\log(xy)$, $\log \frac{x}{y}$, $\log x^k$. Nothing simplifies $\log(x + y)$.

Question 11 ERR-03 · Identifying errors in proofs · spec Err2

A student tries to solve $\sin \theta = \sin 40^\circ$ for $0^\circ \leq \theta < 360^\circ$.

- I Since $\sin \theta = \sin 40^\circ$, we have $\theta = 40^\circ$.
- II The sine function has period 360° , so $\theta = 40^\circ + 360^\circ k$.
- III In the given interval the only solution is $\theta = 40^\circ$.

Which of the following is correct?

Answer: A

'If $\sin \alpha = \sin \beta$ then $\alpha = \beta$ ' is a classic invalid deduction; $\sin(180^\circ - 40^\circ) = \sin 40^\circ$ too.

Full method - how and why

$\sin$ is not one-to-one, so $\sin \theta = \sin \alpha$ has more solutions than $\theta = \alpha$.

- 1 Line I claims $\sin \theta = \sin 40^\circ \Rightarrow \theta = 40^\circ$. This is the error.

why: Sine takes the same value twice per revolution: $\sin(180^\circ - \alpha) = \sin \alpha$.

- 2 So $\theta = 140^\circ$ also satisfies the equation.

why: $\sin 140^\circ = \sin 40^\circ$ - check on the graph or with the symmetry of the unit circle.

- 3 Lines II and III are then correct **statements** in themselves (sine does have period 360°), but they are built on the incomplete line I.

why: The question asks for the **first** error, which is line I.

- 4 The full solution set in $0^\circ \leq \theta < 360^\circ$ is $\theta = 40^\circ$ and $\theta = 140^\circ$.

why: Only two, because the sine is positive in the first and second quadrants.

Where it goes wrong

- Claiming the period of sine is 180° - that is tangent.
- Claiming $\theta = 400^\circ$ was missed - it is outside the stated interval.
- Declaring the work correct because each individual line looks plausible.

Takeaway $\sin \theta = \sin \alpha$ gives $\theta = \alpha + 360^\circ k$ **or** $\theta = 180^\circ - \alpha + 360^\circ k$. **Never cancel a sine.**

Question 12 LGC-10 · The logic of arguments · spec Arg2

For real numbers a and b , let P : ' $a = b$ ' and Q : ' $a^2 = b^2$ '. Which one of the following is true?

Answer: C

$a = b \Rightarrow a^2 = b^2$ (sufficient), but $a = 1, b = -1$ gives $a^2 = b^2$ without $a = b$ (not necessary).

Full method - how and why

The squaring trap again, in its purest form.

- 1 Sufficiency: if $a = b$ then squaring both sides gives $a^2 = b^2$. **True.**

why: Doing the same thing to equal quantities keeps them equal.

- 2 Necessity: does $a^2 = b^2$ force $a = b$? Take $a = 1, b = -1$: $1 = 1$ but $a \neq b$.

why: $a^2 = b^2$ only gives $a = \pm b$.

- 3 So P is sufficient but not necessary.

why: Squaring loses sign information, so it cannot be reversed.

Where it goes wrong

- Treating squaring as reversible - it is exactly this that makes squaring an equation introduce extra solutions.
- Only trying positive values.

Takeaway $a = b \Rightarrow a^2 = b^2$, but $a^2 = b^2 \Rightarrow a = \pm b$. Squaring is a one-way street.

Question 13 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

- I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.
- II A 10% increase followed by a 10% increase is a 21% increase overall.
- III If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Full method - how and why

Three percentage claims. Each is settled by converting to multipliers.

- 1 I: +50% then -50% gives $1.5 \times 0.5 = 0.75$, i.e. 75% of the original. **False.**

why: The decrease acts on the larger amount, so it removes more than the increase added.

- 2 II: +10% twice gives $1.1^2 = 1.21$, a 21% increase. **True.**

why: The extra 1% is the interest on the first year's interest.

- 3 III: $A = 1.25B$ means $B = \frac{A}{1.25} = 0.8A$, which is 20% less than A . **True.**

why: Percentage changes are not symmetric because the base changes.

- 4 II and III only.

why: Match to the options.

Where it goes wrong

- Accepting I because the percentages 'cancel'.
- Rejecting III on the grounds that 25% up should mean 25% down.

Takeaway Always convert to multipliers: percentage changes are asymmetric because the base changes.

Question 14 PRF-02 · Mathematical proof · spec Prf1

Which of the following is a counterexample to the claim '**if n is a positive integer, then $2^n + 1$ is prime**'?

Answer: D

$2^3 + 1 = 9 = 3 \times 3$ is not prime. ($n = 1, 2, 4, 8$ give 3, 5, 17, 257, all prime.)

Full method - how and why

To disprove a universal claim, find one value where the conclusion fails - so test values until one breaks.

- 1 Evaluate $2^n + 1$ for each candidate: $n = 1 \rightarrow 3$, $n = 2 \rightarrow 5$, $n = 3 \rightarrow 9$, $n = 4 \rightarrow 17$, $n = 8 \rightarrow 257$.

why: Systematic evaluation is faster than trying to be clever here.

- 2 $9 = 3 \times 3$ is **not** prime, so $n = 3$ breaks the claim.

why: That is the counterexample: hypothesis satisfied (3 is a positive integer), conclusion false.

- 3 The others - 3, 5, 17, 257 - are all prime, so none of them refutes anything.

why: (257 is prime: it is not divisible by 2, 3, 5, 7, 11, 13, and $17^2 = 289 > 257$.)

Where it goes wrong

- Assuming a large value like $n = 8$ must fail because the numbers get big - size is not the issue.
- Stopping at the first odd value of $2^n + 1$; every one of them is odd, since 2^n is even.

Takeaway Notice the pattern: $2^n + 1$ can only be prime when n is a power of 2 - and 3 is not, which is exactly why it fails.

Question 15 RSN-06 · Chains of reasoning · spec Prf5

Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

Answer: C

36 makes all three statements true, not exactly two. Each of the others makes exactly two true: 16 and 25 (square, < 50), 48 and 12 (multiple of 6, < 50), 144 (square, multiple of 6).

Full method - how and why

Test each candidate against all three statements and count how many are true. You want the one that does **not** give exactly two.

- 1 36: perfect square (6^2) ✓, multiple of 6 ✓, less than 50 ✓ - **three** true.

why: The condition demands exactly two, so 36 is ruled out.

- 2 16: square ✓, multiple of 6 ✗, < 50 ✓ - two. Allowed.

why: Check each of the three statements, do not stop early.

- 3 25: square ✓, multiple of 6 ✗, < 50 ✓ - two. 48: square ✗, multiple of 6 ✓, < 50 ✓ - two. 12: the same pattern - two.

why: All fine.

- 4 144: square (12^2) ✓, multiple of 6 ✓, < 50 ✗ - two. Allowed.

why: So only 36 fails.

Where it goes wrong

- Reading 'exactly two are true' as 'at least two', which would admit 36.
- Missing that $144 = 12^2$ is both a square and a multiple of 6.
- Looking for the value making the fewest statements true instead of the most.

Takeaway 'Exactly k ' excludes $k + 1$ as firmly as $k - 1$; tabulate every statement for every candidate.

Question 16 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Full method - how and why

Inclusion-exclusion: count what you don't want and subtract. Take care with 'less than 1000'.

- 1 The range is 1 to 999, so 999 integers.
why: 'Less than 1000' excludes 1000 itself.
- 2 Multiples of 2: $\lfloor \frac{999}{2} \rfloor = 499$. Multiples of 5: 199. Multiples of both, i.e. of 10: 99.
why: Divide and round down. 10 is the LCM of 2 and 5.
- 3 Divisible by 2 or 5: $499 + 199 - 99 = 599$.
why: Subtract the overlap, which was counted twice.
- 4 Neither: $999 - 599 = 400$.
why: Sanity check: in each block of 10 consecutive integers exactly 4 (ending in 1, 3, 7, 9) survive, and $999 = 99$ full blocks plus 991-999, giving $99 \times 4 + 4 = 400$ ✓.

Where it goes wrong

- Using 1000 as the count and getting 400 by luck with wrong intermediate figures - or getting 399 by an off-by-one at the top.
- Forgetting to subtract the multiples of 10, giving $999 - 698 = 301$.

Takeaway $|A \cup B| = |A| + |B| - |A \cap B|$, with the intersection counted via the LCM; then subtract from the total.

Question 17 QUA-06 · Quadratic functions · spec MM1.3

The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

- I $f(x) > 0$ for all real x
- II The graph of $y = f(x)$ is symmetric about the line $x = 3$
- III The equation $f(x) = 4$ has exactly one real solution

Answer: H

$f(x) = (x - 3)^2 + 4 \geq 4 > 0$, so I is true. The axis of symmetry is $x = 3$, so II is true. $f(x) = 4$ only when $(x - 3)^2 = 0$, i.e. $x = 3$: exactly one solution, so III is true.

Full method - how and why

Three statements about one quadratic. Completing the square once answers all three at a stroke.

- 1 Complete the square: $x^2 - 6x + 13 = (x - 3)^2 + 4$.

why: This single form contains the range, the line of symmetry and the vertex - so do it first, before looking at the statements.

- 2 I: $(x - 3)^2 \geq 0$ so $f(x) \geq 4 > 0$ for every x . **True.**

why: The minimum value is 4, comfortably positive; equivalently the discriminant $36 - 52 < 0$, so the curve never reaches the axis.

- 3 II: the vertex is at $x = 3$, and a parabola is symmetric about the vertical line through its vertex. **True.**

why: Symmetry about $x = h$ is immediate from the $(x - h)^2$ form.

- 4 III: $f(x) = 4$ means $(x - 3)^2 = 0$, so $x = 3$ only - exactly one solution. **True.**

why: A horizontal line through the vertex meets the parabola once; any higher line meets it twice.

- 5 All three hold, so the answer is 'I, II and III'.

why: Decide each statement independently, then match to the option list.

Where it goes wrong

- Assuming $f(x) = 4$ has two solutions because quadratics 'usually' do - $y = 4$ is the tangent line at the vertex.
- Reading the symmetry line off as $x = -3$.

Takeaway Completing the square answers questions about sign, range, symmetry and repeated roots all at once.

Question 18 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n+1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$). No two consecutive positive integers are both squares, so the answer is 0.

Full method - how and why

A 'for how many' question that is really about irrationality. The answer is 0, and seeing why needs one sharp observation about consecutive integers.

- 1 Suppose $\sqrt{n} + \sqrt{n+1} = r$ is rational and square it: $2n+1 + 2\sqrt{n(n+1)} = r^2$.
why: Squaring is the standard way to get at a sum of surds; the rational parts separate from the single remaining surd.
- 2 Rearrange: $\sqrt{n(n+1)} = \frac{r^2 - 2n - 1}{2}$, so $n(n+1)$ would have to be a perfect square.
why: A rational square root of an integer is an integer, so $n(n+1)$ must be a perfect square.
- 3 But $n^2 < n(n+1) < (n+1)^2$ for every $n \geq 1$, so $n(n+1)$ lies strictly between consecutive squares.
why: There is no integer square strictly between n^2 and $(n+1)^2$ - this is the contradiction.
- 4 Hence no n in $1 \leq n \leq 100$ works: the answer is 0.
why: The bound 100 is a red herring; the argument rules out every positive integer.

Where it goes wrong

- Testing $n = 3$ ($\sqrt{3} + 2$) and concluding 'sometimes it works' because one of the two roots is nice.
- Answering 9 or 10 by counting perfect squares up to 100 - that would answer a different question.

Takeaway $n(n+1)$ is never a perfect square for $n \geq 1$, because it is squeezed strictly between n^2 and $(n+1)^2$.

Question 19 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.

II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.

III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).

Full method - how and why

A subtle one: the **claim is true** but one line asserts more than was proved. Distinguish a false conclusion from an unjustified step.

1 Check line I: $f'(x) = 3x^2 - 6x + 3 = 3(x^2 - 2x + 1) = 3(x - 1)^2$. Correct.

why: Differentiate term by term, then factorise.

2 Check line II: a square is non-negative, so $f'(x) \geq 0$, with equality only at $x = 1$. Correct.

why: The word 'only' is right - $(x - 1)^2 = 0$ just at $x = 1$.

3 Line III says ' $f'(x) > 0$ for all x ' - **false at $x = 1$ **, where $f'(1) = 0$. So line III overstates line II.

why: It asserts a strict inequality that was never established.

4 Is the claim itself true? Yes: $f' \geq 0$ everywhere with equality at a single isolated point still gives $f(a) < f(b)$ for $a < b$.

why: A derivative vanishing at one point does not create a flat stretch; there is no turning point, because f' does not change sign.

Where it goes wrong

- Concluding the claim is false because $f'(1) = 0$ - a stationary point of inflexion is not a turning point.
- Blaming line I or II, both of which are correct.
- Calling the proof completely correct, ignoring the strictness error in line III.

Takeaway $f' \geq 0$ with equality only at isolated points still means strictly increasing - but a proof must not claim strict positivity it has not shown.

Question 20 LGC-15 · The logic of arguments · spec Arg2

Let P : 'the integer n is divisible by 4' and Q : 'the integer n^2 is divisible by 8'. Which one of the following is true?

Answer: A

If $4 \mid n$ then $16 \mid n^2$, so $8 \mid n^2$: P is sufficient. Conversely, if $8 \mid n^2$ then n is even, say $n = 2m$, and $8 \mid 4m^2$ forces m to be even, so $4 \mid n$: P is also necessary. (Tempting wrong answer: 'sufficient but not necessary' - but no counterexample exists.)

Full method - how and why

A genuine 'necessary **and** sufficient' case, dressed to look like the usual 'sufficient only'. You must prove both directions rather than assume a counterexample exists.

1 Sufficiency: if $4 \mid n$, write $n = 4k$. Then $n^2 = 16k^2$, which is divisible by 8. **True.**

why: 16 is a multiple of 8, so divisibility by 8 follows immediately.

2 Necessity: suppose $8 \mid n^2$. Then n^2 is even, so n is even - write $n = 2m$.

why: An odd number squared is odd, so n cannot be odd.

3 Then $n^2 = 4m^2$, and $8 \mid 4m^2$ forces $2 \mid m^2$, so m is even, say $m = 2j$.

why: Divide the divisibility through by the common factor 4.

4 So $n = 4j$, i.e. $4 \mid n$. Both directions hold: necessary **and** sufficient.

why: No counterexample exists, which is what distinguishes this from the usual pattern.

Where it goes wrong

- Answering 'sufficient but not necessary' out of habit, without actually hunting for a counterexample. Test $n = 2$: $n^2 = 4$, not divisible by 8 - so it is not a counterexample; $n = 6$: 36, not divisible by 8. The hunt keeps failing, which is the clue.
- Assuming $8 \mid n^2 \Rightarrow 8 \mid n$ - false ($n = 4$ gives $n^2 = 16$).

Takeaway Before declaring a condition 'not necessary', actually look for a counterexample; repeated failure to find one is a signal to prove the converse instead.